Lecture 22: Binary Decision Task and ROC

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🔑 Learning Objectives

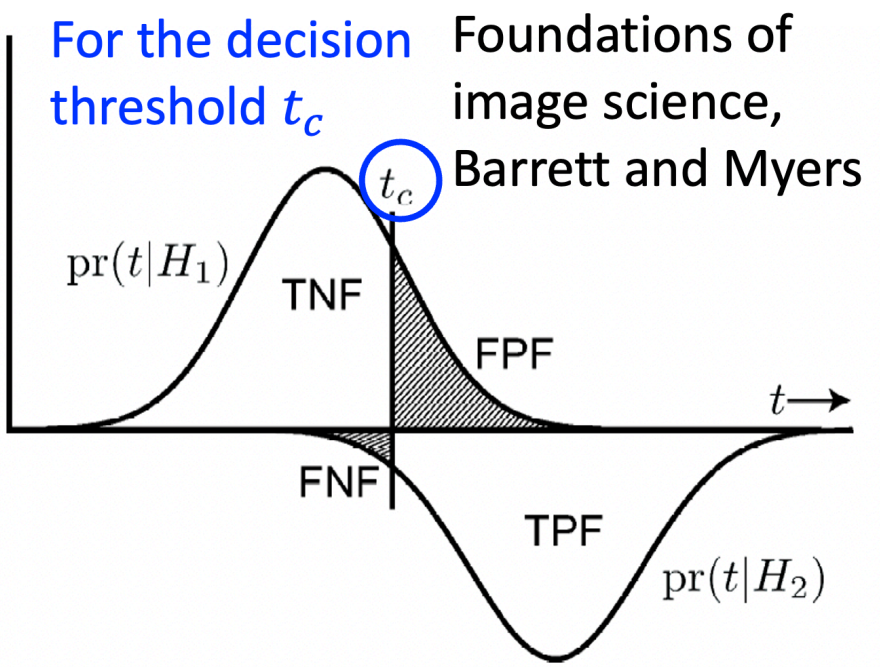
By the end of this lecture, you should be able to understand:

- Receiver operating characteristic (ROC) curves and area under the ROC curve (AUC)
- Likelihood ratio

Binary Decision Outcomes

In a binary decision task, there are four possible outcomes based on the true state of nature and the observer's decision:

- D_2 and H_2 → True positive (TP)
- D_2 and H_1 → False positive (FP): *Type I error*
- D_1 and H_2 → False negative (FN): *Type II error*
- D_1 and H_1 → True negative (TN)



Errors

- **Type I**: The signal is absent (H_1), but the observer decides it is present (D_2). → This may lead to **unnecessary** *testing* or *biopy*.
- **Type II**: The signal is present (H_2), but the observer decides it is absent (D_1). → A lesion may be **missed** during screening.

Sensitivity and Specificity

Let the following denote the number of decisions made under each outcome:

- N_{TP}: Number of **true positive** decisions (H_2 and D_2)
- N_{FP}: Number of **false positive** decisions (H_1 and D_2)
- N_{TN}: Number of **true negative** decisions (H_1 and D_1)
- N_{FN}: Number of **false negative** decisions (H_2 and D_1)
- $N = N_{TP} + N_{FP} + N_{TN} + N_{FN}$: Total number of decisions made by an observer

True- and False-Positive and -Negative Fractions

- True positive fraction (TPF), also called ***sensitivity***:

$$\mathrm{TPF} = \Pr(D_2|H_2) = \frac{N_{TP}}{N_{TP} + N_{FN}} = \Pr(t \geq t_c|H_2) = \int_{t_c}^{\infty} dt\, \mathrm{pdf}(t|H_2)$$

- False positive fraction (FPF):

$$\mathrm{FNF} = \Pr(D_1|H_2) = \frac{N_{FN}}{N_{TP} + N_{FN}} = \Pr(t < t_c|H_2) = \int_{-\infty}^{t_c} dt\, \mathrm{pdf}(t|H_2)$$

- Flase negative fraction (FNF):

$$\mathrm{FPF} = \Pr(D_2|H_1) = \frac{N_{FP}}{N_{TP}+N_{FP}} = \Pr(t \geq t_c|H_1) = \int_{t_c}^{\infty} dt\ \mathrm{pdf}(t|H_1)$$

- True negative fraction (TNF), also called ***specificity***:

$$\mathrm{TNF} = \Pr(D_1|H_1) = \frac{N_{TN}}{N_{TN}+N_{FP}} = \Pr(t < t_c|H_1) = \int_{-\infty}^{t_c} dt\ \mathrm{pdf}(t|H_1)$$

- $\mathrm{TPF} + \mathrm{FNF} = 1$
- $\mathrm{FPF} + \mathrm{TNF} = 1$

🧠 *Recall continuous random variables* (in Lecture 18):
$\Pr(a \leq x \leq b) = \int_a^b dx'\ \mathrm{pdf}(x'), \int_{-\infty}^{\infty} dx'\ \mathrm{pdf}(x') = 1$

Performance Measures for Binary Tasks

Various metrics allow for quantitative comparison of imaging systems and observer performance.

Accuracy

$$\mathrm{accuracy} = \lim_{N\to\infty} \frac{N_{TP}+N_{TN}}{N}$$

Suppose you are screening patients for a rare disease that affects only 0.1% of the population, i.e., $\Pr(H_2) = 0.0001$. By classifying every patient as normal (D_1), you achieve 99.9% accuracy.

However, this misclassifies every actual case of disease (H_2) as normal, yielding:

- Type II error: $\mathrm{FNF} = 1$
- Sensitivity: $\mathrm{TPF} = 0$

Accuracy can mislead when class distributions are highly imbalanced. A test may appear effective even though it fails to detect rare but important cases.

Limitation: Accuracy is heavily influenced by prevalence, i.e., the prior probabilities $\Pr(H_1)$ and $\Pr(H_2)$.

Positive and Negative Predictive Values

Another set of performance metrics are the **positive predictive value (PPV)** and **negative predictive value (NPV)**. These quantify the probability that a given decision corresponds to the true condition.

$$\mathrm{PPV} = \Pr\left(H_2 | D_2\right) = \lim_{N\to\infty} \frac{N_{TP}}{N_{TP} + N_{FP}}$$

$$\mathrm{NPV} = \Pr\left(H_1 | D_1\right) = \lim_{N\to\infty} \frac{N_{TN}}{N_{TN} + N_{FN}}$$

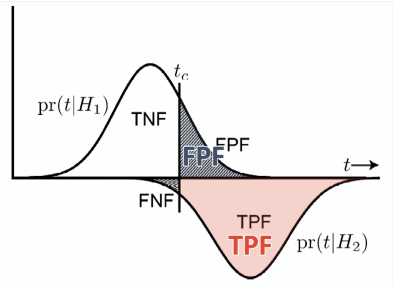
Receiver Operating Characteristic (ROC) Curve

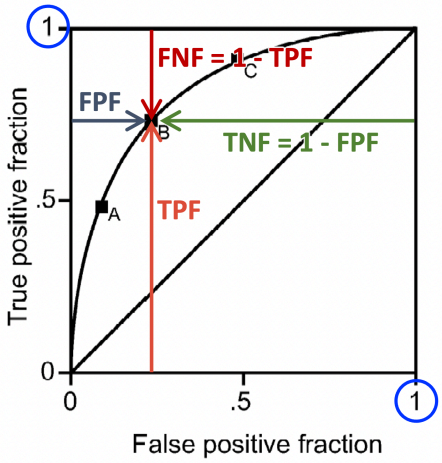
Each pair TPF-FPF corresponds to a specific decision threshold t_c.

Since:

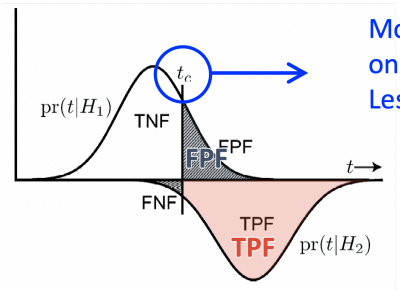
- $\mathrm{TPF} + \mathrm{FNF} = 1$
- $\mathrm{FPF} + \mathrm{TNF} = 1$

a point on the ROC curve captures all four outcome fractions as a function of t_c.

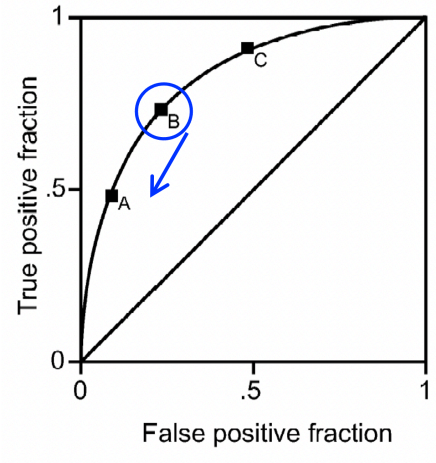




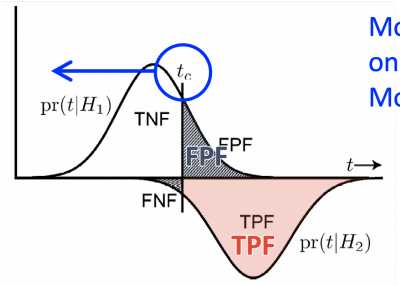
Foundations of image science, Barrett and Myers



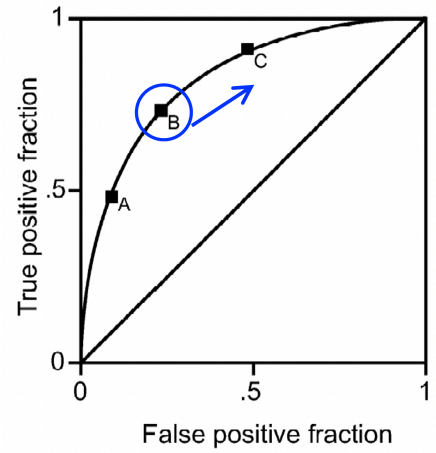
More **strict** threshold
on positive decision D_2
Less number of D_2



Foundations of image science, Barrett and Myers



More **relaxed** threshold
on positive decision D_2
More number of D_2



Foundations of image science, Barrett and Myers

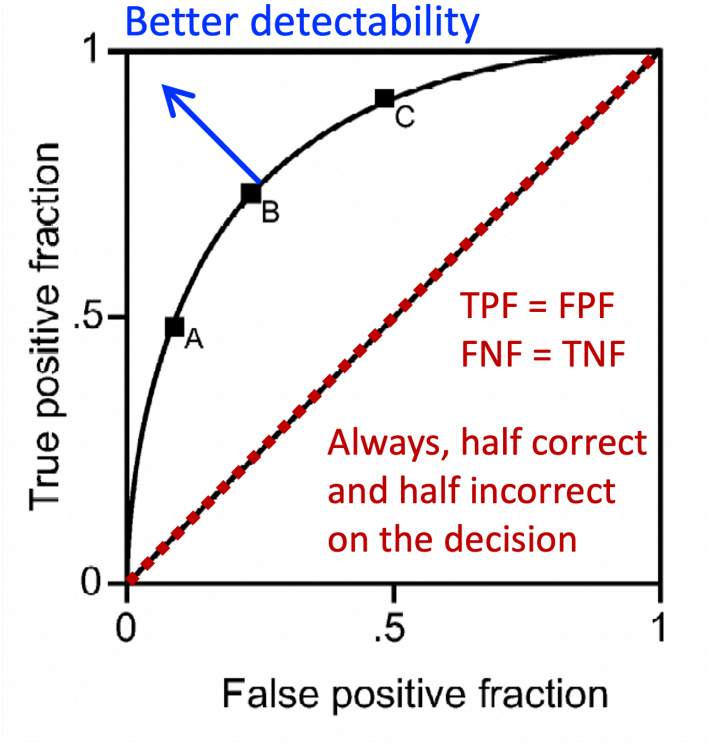
Assume the decision variable t is normally distributed: $t_j \sim \mathcal{N}\left(E_j[t], \sigma_j^2\right)$

Then:

$$\mathrm{TPF} = \frac{1}{2}\left[1 - \mathrm{erf}\left(\frac{t_c - \mathrm{E}[t_2]}{\sqrt{2\sigma_2^2}}\right)\right]$$

$$\mathrm{FPF} = \frac{1}{2}\left[1 - \mathrm{erf}\left(\frac{t_c - \mathrm{E}[t_1]}{\sqrt{2\sigma_1^2}}\right)\right]$$

- Gauss error function: $\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z dy\ \exp[-y^2]$

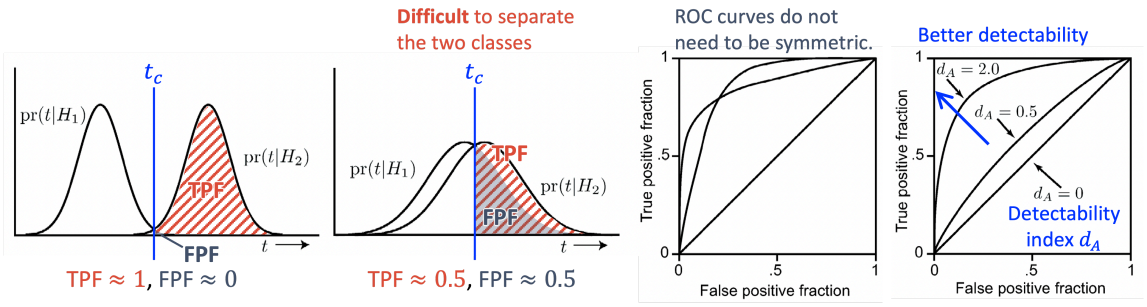


Foundations of image science, Barrett and Myers

The ROC curve, summarizes:

- Task difficulty
- Performance of the decision strategy, and
- Data quality for enabling the observer to perform the specified task

Unlike accuracy, ROC-based metrics are **independent of prevalence**, making them more robust and widely applicable.



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Area Under the ROC Curve (AUC)

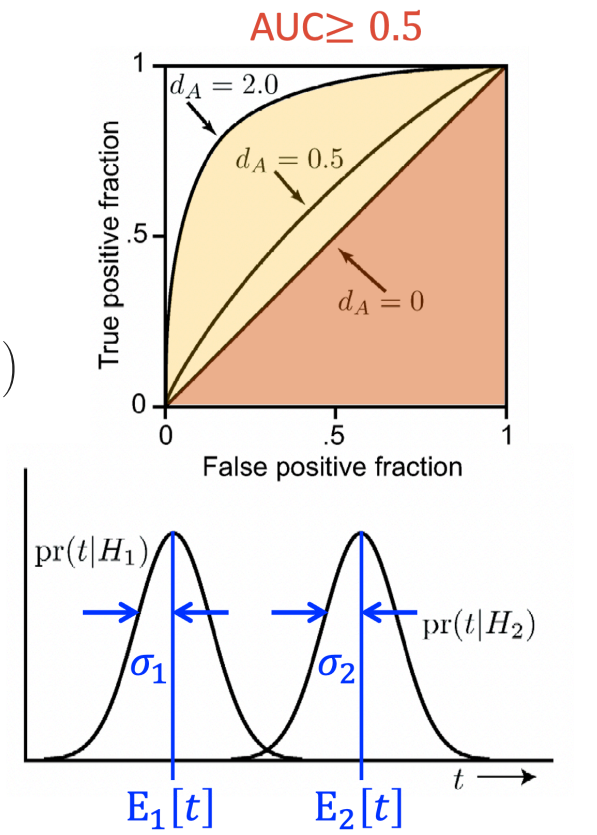
$$\text{AUC} = \int_0^1 d\text{FPF}\ \text{TPF}(\text{FPF})$$

If t is normally distributed under each hypothesis, $t \sim \mathcal{N}\left(\mathrm{E}_j[t], \sigma_j^2\right)$, the **signal-to-noise ratio (SNR) for** t is:

$$\mathrm{SNR}_t = \frac{\mathrm{E}_2[t] - \mathrm{E}_1[t]}{\sqrt{\frac{1}{2}\sigma_1^2 + \frac{1}{2}\sigma_2^2}} \qquad \mathrm{AUC} = \frac{1}{2} + \frac{1}{2}\mathrm{erf}\left(\frac{\mathrm{SNR}_t}{2}\right)$$

Detectability index d_A

$$d_A = 2\mathrm{erf}^{-1}[2(\mathrm{AUC}) - 1]$$



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Likelihood Ratio

Minimum Average Cost

A decision strategy that *minimizes expected cost* is known as Bayes decision theory. The overall average cost, also called **Bayes risk**, is given by (binary case, $L = 2$):

$$\sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr\left(H_j, D_i\right)$$

$$= \sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr\left(H_j|D_i\right)\Pr\left(D_i\right) = \sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr\left(D_i|H_j\right)\Pr\left(H_j\right)$$

$$= C_{11}\Pr\left(D_1|H_1\right)\Pr\left(H_1\right) + C_{12}\Pr\left(D_1|H_2\right)\Pr\left(H_2\right)$$
$$+ C_{21}\Pr\left(D_2|H_1\right)\Pr\left(H_1\right) + C_{22}\Pr\left(D_2|H_2\right)\Pr\left(H_2\right)$$

With:

$$\Pr\left(D_i|H_j\right) = \int_{\Gamma_i} d^M g\, \mathrm{pdf}\left(\mathbf{g}|H_j\right)$$

Bayes risk becomes:

$$\sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr(H_j, D_i)$$

$$= C_{11}\Pr(H_1)\int_{\Gamma_1} d^M g\,\mathrm{pdf}(\mathbf{g}|H_1) + C_{12}\Pr(H_2)\int_{\Gamma_1} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2)$$

$$+ C_{21}\Pr(H_1)\int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_1) + C_{22}\Pr(H_2)\int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2)$$

With:

$$\int_{\Gamma_1} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2) + \int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2) = 1$$

Bayes risk becomes:

$$\sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr(H_j, D_i)$$

$$= C_{11}\Pr(H_1)\left[1 - \int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_1)\right] + C_{12}\Pr(H_2)\left[1 - \int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2)\right]$$

$$+ C_{21}\Pr(H_1)\int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_1) + C_{22}\Pr(H_2)\int_{\Gamma_2} d^M g\,\mathrm{pdf}(\mathbf{g}|H_2)$$

$$\sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr(H_j, D_i)$$

$$= C_{11}\Pr(H_1) + C_{12}\Pr(H_2)$$

$$+ \int_{\Gamma_2} d^M g\,\left[(C_{21} - C_{11})\Pr(H_1)\mathrm{pdf}(\mathbf{g}|H_1) - (C_{12} - C_{22})\Pr(H_2)\mathrm{pdf}(\mathbf{g}|H_2)\right]$$

- $\scriptsize{C_{11} \text{Pr}(H_{1})
- C_{12} \text{Pr}(H_{2})}$: Constant baseline risk
- $\scriptsize{\int_{\Gamma_{2}}d^{M}g, [(C_{21} - C_{11}) \text{Pr}(H_{1}) \text{pdf}(\mathbf{g}|H_{1})
- (C_{12} - C_{22}) \text{Pr}(H_{2}) \text{pdf}(\mathbf{g}|H_{2})]}$: $Riskdependingonthedecisionregion\Gamma_{2}$

To minimize Bayes risk, we should select Γ_2 such that:

$$(C_{12} - C_{22})\Pr(H_2)\mathrm{pdf}(\mathbf{g}|H_2) > (C_{21} - C_{11})\Pr(H_1)\mathrm{pdf}(\mathbf{g}|H_1)$$

Assume $(C_{21} - C_{11}) > 0$ and $(C_{12} - C_{22}) > 0$. This is typically satisfied if the cost for incorrect decisions is greater than that for correct ones, e.g., under a squared error loss $(\hat{\theta} - \theta)^2$.

$$\frac{(C_{12}-C_{22})\Pr(H_2)\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)} > (C_{21}-C_{11})\Pr(H_1)$$

$$\frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)} > \frac{(C_{21}-C_{11})\Pr(H_1)}{(C_{12}-C_{22})\Pr(H_2)}$$

The quantity $\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)}$ is called the **likelihood ratio**. The decision rule becomes:

$$\Lambda(\mathbf{g}) \underset{D_1}{\overset{D_2}{\gtrless}} \frac{(C_{12}-C_{22})\ \Pr(H_2)}{(C_{21}-C_{11})\ \Pr(H_1)}.$$

- $\frac{(C_{12}-C_{22})\ \Pr(H_2)}{(C_{21}-C_{11})\ \Pr(H_1)}$: Bayes criterion

🧠 *The conditional PDF, often referred to as the **likelihood** functions, characterizes how the data $\mathbf{g}$ are distributed given a specific underlying hypothesis H_j.* (in Lecture 21)

Minimum-Error Detector

The probability of classification error, denoted as P_e, is the total probability of making an *incorrect* decision:

$$\begin{aligned} P_e &= \Pr(H_1, D_2) + \Pr(H_2, D_1) \\ &= \Pr(D_2|H_1)\Pr(H_1) + \Pr(D_1|H_2)\Pr(H_2) \end{aligned}$$

This can be compared with the general expression for Bayes risk (average decision cost):

$$\begin{aligned} &\sum_{i=1}^{L}\sum_{j=1}^{L} C_{ij}\Pr(H_j, D_i) \\ &= C_{11}\Pr(D_1|H_1)\Pr(H_1) + C_{12}\Pr(D_1|H_2)\Pr(H_2) \\ &+ C_{21}\Pr(D_2|H_1)\Pr(H_1) + C_{22}\Pr(D_2|H_2)\Pr(H_2) \end{aligned}$$

To minimize this risk, the **likelihood ratio** is used:

$$\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)} > \frac{(C_{21}-C_{11})\Pr(H_1)}{(C_{12}-C_{22})\Pr(H_2)}$$

Minimizing P_e is equivalent to minimizing the Bayes risk when

- $C_{11} = C_{22} = 0$: No cost for correct decisions
- $C_{12} = C_{21}$: Equal cost for both types of errors

In this case, the decision rule simplifies to:

$$\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)} \underset{D_1}{\overset{D_2}{\gtrless}} \frac{\Pr(H_1)}{\Pr(H_2)}$$

Or, equivalently:

$$\mathrm{pdf}\big(\mathbf{g}|H_2\big)\Pr\big(H_2\big) \underset{D_1}{\overset{D_2}{\gtrless}} \mathrm{pdf}\big(\mathbf{g}|H_1\big)\Pr\big(H_1\big)$$

Since both $\Pr(H_j) > 0$ and $\mathrm{pdf}(\mathbf{g}|H_j) > 0$, we can apply Bayes' theorem:

$$\Pr(H_j|\mathbf{g}) = \frac{\mathrm{pdf}(\mathbf{g}|H_j)\Pr(H_j)}{\sum_j \mathrm{pdf}(\mathbf{g}|H_j)\Pr(H_j)} = \frac{\mathrm{pdf}(\mathbf{g}|H_j)\Pr(H_j)}{\mathrm{pdf}(\mathbf{g})}$$

$$\mathrm{pdf}(\mathbf{g}|H_j)\Pr(H_j) = \Pr(H_j|\mathbf{g})\mathrm{pdf}(\mathbf{g})$$

- Prior: $\Pr(H_j)$
- Likelihood: $\mathrm{pdf}(\mathbf{g}|H_j)$
- Posteriori: $\Pr(H_j|\mathbf{g})$

Substituting back into the inequality:

$$\mathrm{pdf}\big(\mathbf{g}|H_2\big)\Pr\big(H_2\big) \underset{D_1}{\overset{D_2}{\gtrless}} \mathrm{pdf}\big(\mathbf{g}|H_1\big)\Pr\big(H_1\big)$$

$$\Pr(H_2|\mathbf{g})\mathrm{pdf}(\mathbf{g}) \underset{D_1}{\overset{D_2}{\gtrless}} \Pr(H_1|\mathbf{g})\mathrm{pdf}(\mathbf{g})$$

$$\Pr(H_2|\mathbf{g}) \underset{D_1}{\overset{D_2}{\gtrless}} \Pr(H_1|\mathbf{g})$$

This is the **minimum-error criterion**, also known as the **maximum a posteriori probability (MAP)** decision rule. The observer chooses the hypothesis with the highest posterior probability given the data.

Maximum-Likelihood Criterion

When the observer has no prior knowledge of the hypothesis probabilities, it is common to assume: $\Pr(H_1) = \Pr(H_2)$

Under this assumption, the MAP decision rule reduces to the **maximum-likelihood (ML)** criterion:

$$\Lambda(\mathbf{g}) = \frac{\mathrm{pdf}\big(\mathbf{g}|H_2\big)}{\mathrm{pdf}\big(\mathbf{g}|H_1\big)} \underset{D_1}{\overset{D_2}{\gtrless}} \frac{\Pr\big(H_1\big)}{\Pr\big(H_2\big)} = 1$$

$$\text{Choose } H_2 \text{ if } \mathrm{pdf}\big(\mathbf{g}|H_2\big) > \mathrm{pdf}\big(\mathbf{g}|H_1\big)$$